# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is a ray infinite in one direction? Student says: No

**Think about the definition of a ray.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **A ray has one endpoint and extends forever in one direction. The answer is yes.** | **Yes** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Does a line extend infinitely in both directions? Student says: No

**Think about the definition of a line.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **A line has no endpoints and continues forever in both directions.** | **Yes** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

How many points are needed to define a unique line? Student's answer: 1

**Think about how many points fix a line.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **Two distinct points determine exactly one line.** | **2** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

How many endpoints does a line segment have? Student's answer: 1

**Think about the definition of a segment.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **A segment has two endpoints, one at each end.** | **2** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

At what angle do perpendicular lines meet? Student's answer: 45°

**Recall the meaning of perpendicular.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **Perpendicular lines meet at a right angle, which is 90°.** | **90°** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

An angle bisector divides an angle into how many congruent angles? Student's answer: 3

**Think about what a bisector does.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **A bisector cuts an angle into two congruent angles.** | **2** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A reflection maps segment AB to AB. Is AB congruent to AB? Student says: No

**Think about whether a reflection changes length.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **Reflections are rigid motions. They preserve length, so the segments are congruent.** | **Yes** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A 40° angle is rotated about a point. What is the measure of the image angle? Student's answer: 80°

**Think about what a rotation preserves.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **Rotations preserve angle measure. The image angle is still 40°.** | **40°** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Does a dilation preserve distances? Student says: Yes

**Think about whether a dilation changes size.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **A dilation with scale factor not equal to 1 changes distances, so it is not a rigid motion.** | **No** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Does a 90° rotation preserve area? Student says: No

**Think about rigid motions.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **Rotations are rigid motions, so they preserve area.** | **Yes** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Does a reflection preserve orientation? Student says: Yes

**Think about whether the figure is flipped.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **A reflection flips the figure, so it reverses orientation.** | **No** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

he point (3, 4) is rotated 90° counterclockwise about the origin. What is the image? Student's answer: (4, -3

**Use the rule (x, y) (-y, x).**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The student used the rule for a clockwise rotation. The counterclockwise image is (-4, 3).** | **(-4, 3)** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ABC  DEF, and A = 50°. What is D? Student's answer: 130°

**Use corresponding parts.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **Corresponding angles of congruent triangles are congruent. D = 50°.** | **50°** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

PQR  XYZ, and PQ = 7. What is XY? Student's answer: 14

**Use corresponding parts.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The student doubled the length. Corresponding sides are equal, so XY = 7.** | **7** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

In ABC DEF with A D, B E, and C F, which side corresponds to AC? Student's answer: DE

**Match the vertices in order.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **AC uses the vertices A and C, which match D and F. The corresponding side is DF.** | **DF** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

If ABC  DEF, is C  F? Student says: No

**Use the correspondence.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **C corresponds to F, so the angles are congruent by CPCTC.** | **Yes** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

If ABC  DEF, BC = 10, and EF = 12, are BC and EF congruent? Student says: Yes

**Compare the lengths.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The lengths are 10 and 12, which are not equal. Corresponding sides of congruent triangles must have equal lengths, so these triangles are not congruent as stated.** | **No** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

es have an angle of 60° opposite a side of length 9. What is the corresponding angle in the other triangle?

**Use CPCTC.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **Corresponding angles are congruent, so the corresponding angle is 60°.** | **60°** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Two triangles have sides 3, 4, and 5 in both. Are they congruent? Student says: No

**Check which criterion applies.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **All three pairs of sides are congruent, so the triangles are congruent by SSS.** | **Yes, by SSS** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

angles 50° and 60° with a side of 4 between those angles in both. Which criterion proves congruence? Stu

**Look at where the side is located.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The side is between the two angles, so the criterion is ASA. AAA does not prove congruence.** | **ASA** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is SSA a valid criterion for proving triangles congruent? Student says: Yes

**Think about whether the triangle is determined.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **SSA can describe two different triangles, so it is not a valid congruence criterion.** | **No** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ght triangles have equal hypotenuses and one pair of equal legs. Which criterion applies? Student's answe

**Think about criteria for right triangles.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **For right triangles, the correct criterion is HL, for hypotenuse and leg.** | **HL** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Two triangles have all three angles equal. Are they congruent? Student says: Yes

**Think about size.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **AAA shows the triangles are similar, not congruent. Their sides can have different lengths.** | **No, only similar** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

angles have sides 4 and 6 with a 40° angle between them in both. Which criterion applies? Student's answ

**Check what is given.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The given parts are two sides and the included angle, so the criterion is SAS.** | **SAS** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Two angles are vertical angles. One measures 70°. What is the other? Student's answer: 110°

**Recall the vertical angle theorem.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **Vertical angles are congruent. The other angle is 70°.** | **70°** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

parallel lines are cut by a transversal. An alternate interior angle is 65°. What is the other? Student's answe

**Recall the alternate interior angle theorem.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **Alternate interior angles are congruent, so the other angle is 65°.** | **65°** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A point is on the perpendicular bisector of a segment. Is it equidistant from the endpoints? Student says: No

**Think about the perpendicular bisector theorem.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **Every point on the perpendicular bisector is equidistant from the endpoints.** | **Yes** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Two adjacent angles form a straight line. One is x and the other is 2x. What is x? Student's answer: 90°

**Angles on a straight line add to 180°.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **x + 2x = 180 gives x = 60°. The student used 180 ÷ 2 for the wrong pair.** | **60°** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

An angle measures 80°, and its bisector is drawn. What is each of the two new angles? Student's answer: 80

**Divide the angle into two equal parts.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The bisector divides 80° into two equal angles of 40° each.** | **40°** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Corresponding angles formed by a transversal are both 75°. Are the lines parallel? Student says: No

**Recall the converse of the corresponding angles theorem.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **Congruent corresponding angles prove the lines are parallel.** | **Yes** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A triangle has angles of 40° and 75°. What is the third angle? Student's answer: 75°

**Angles in a triangle add to 180°.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The student copied one angle. 180 - (40 + 75) = 65°.** | **65°** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A triangle has a base of 12. What is the length of its midsegment parallel to the base? Student's answer: 12

**Recall the midsegment theorem.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The midsegment parallel to the third side is half that side. Its length is 6.** | **6** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ngle of a triangle is 115°. One remote interior angle is 50°. What is the other remote interior angle? Student's

**An exterior angle equals the sum of the two remote interior angles.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The student added the angles. 115 - 50 = 65°.** | **65°** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

An isosceles triangle has base angles of 70°. What is the vertex angle? Student's answer: 110°

**Use the sum of angles in a triangle.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The student subtracted one base angle only. 180 - 2(70) = 40°.** | **40°** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Can a triangle have sides of 4, 6, and 11? Student says: Yes

**Check the triangle inequality.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **4 + 6 = 10, which is less than 11. These sides cannot form a triangle.** | **No** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

triangle with sides 3, 5, and 6, which angle is largest? Student's answer: the angle opposite the side of leng

**The largest angle is opposite the longest side.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The longest side is 6, so the largest angle is opposite it.** | **The angle opposite the side of length 6** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

In a parallelogram, one angle is 110°. What is the opposite angle? Student's answer: 70°

**Recall the opposite angles theorem.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **Opposite angles of a parallelogram are congruent. The opposite angle is 110°.** | **110°** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

The diagonals of a parallelogram meet at O. If AC = 14, what is AO? Student's answer: 14

**Recall that diagonals bisect each other.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The diagonals bisect each other, so AO is half of AC. AO = 7.** | **7** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

In a parallelogram, one angle is 110°. What is an adjacent angle? Student's answer: 110°

**Consecutive angles are supplementary.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **Consecutive angles of a parallelogram add to 180°. The adjacent angle is 70°.** | **70°** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A parallelogram has sides 5 and 8. What is its perimeter? Student's answer: 13

**Opposite sides are equal.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The student counted each side once. The perimeter is 2(5 + 8) = 26.** | **26** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A quadrilateral has both pairs of opposite sides congruent. Is it a parallelogram? Student says: No

**Check the parallelogram criteria.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **A quadrilateral with both pairs of opposite sides congruent is a parallelogram.** | **Yes** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

In a rhombus, what is the angle between the diagonals? Student's answer: 45°

**Recall the properties of a rhombus.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The diagonals of a rhombus are perpendicular, so they meet at 90°.** | **90°** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

dilation with scale factor 3 maps a side of length 4 to a new side. What is the new length? Student's answer

**Multiply by the scale factor.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The student added 3 to 4. Multiply: 4 × 3 = 12.** | **12** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Are two triangles similar if two pairs of angles are congruent? Student says: No

**Recall the AA similarity criterion.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **Two pairs of congruent angles guarantee similarity by AA.** | **Yes** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

th 3 in the small triangle matches a side of 6 in the large triangle. A side of 4 in the small triangle is what in

**Use the scale factor.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The scale factor is 6 ÷ 3 = 2. So 4 × 2 = 8.** | **8** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is a dilation followed by a translation a similarity transformation? Student says: No

**Think about what a similarity transformation allows.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **Dilations and translations both preserve shape, and combining them keeps similarity.** | **Yes** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A dilation has scale factor 1/2. A side of length 10 is dilated. What is the new side? Student's answer: 20

**Multiply by the scale factor.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **A scale factor of 1/2 makes the side smaller. 10 × 1/2 = 5.** | **5** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A dilation has scale factor 2/3. What is the ratio of the areas? Student's answer: 2/3

**Think about how area scales.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **Area scales by the square of the scale factor. $(2/3)^2 = 4/9$.** | **4/9** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Similar triangles have AB = 4, DE = 6, and BC = 8. What is EF if EF corresponds to BC? Student's answer: 10

**Set up a proportion.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **4/6 = 8/x, so x = 12. The student added the sides.** | **12** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A line parallel to BC crosses AB at D and AC at E. AD = 3, DB = 6, and AE = 4. What is EC? Student's answer: 2

**Use the side splitter theorem.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **AD/DB = AE/EC, so 3/6 = 4/EC. Then EC = 8.** | **8** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

t triangle, the altitude to the hypotenuse splits it into segments of 4 and 9. What is the altitude? Student's a

**Use the geometric mean.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The altitude is the geometric mean of the segments: $\sqrt{(4 \times 9)} = 6$.** | **6** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

on casts a 4-foot shadow. A tree casts a 20-foot shadow at the same time. How tall is the tree? Student's a

**Set up a proportion with the shadows.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The ratio is 6/4 = height/20, so height = 30 feet.** | **30 feet** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

sides 4 and 6 with an equal included angle, and sides 6 and 9 with the same included angle. Are they simil

**Check the SAS similarity criterion.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **4/6 = 6/9 = 2/3 and the included angles are equal, so the triangles are similar by SAS.** | **Yes** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

of a triangle is 9 and is divided by the centroid in a 2:1 ratio. What is the longer part from the vertex? Studer

**Think about how the centroid divides a median.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The longer part is 2/3 of the median, so 2/3 × 9 = 6.** | **6** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

les have two sides and the included angle congruent. Which method proves congruence? Student's answe

**Check where the angle is located.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The angle is between the two sides, so the correct criterion is SAS.** | **SAS** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ave sides 5 and 10 matching, and a side of 14 in the second triangle matches a side of x in the first. Find x. S

**Set up a proportion.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **5/10 = x/14, so x = 7.** | **7** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

In similar triangles, AB = 3 and DE = 6, and AC = 5. What is DF? Student's answer: 5

**Use the scale factor.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The scale factor is 6 ÷ 3 = 2. DF = 2 × 5 = 10.** | **10** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

s a 2-meter shadow. A building casts a 14-meter shadow at the same time. How tall is the building? Studer

**Set up a proportion.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The ratio is 3/2 = height/14, so height = 21 meters.** | **21 meters** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Triangle A has sides 5, 12, 13 and triangle B has sides 10, 24, 26. Are they similar? Student says: No

**Compare the ratios of corresponding sides.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **Each side in B is twice the matching side in A, so they are similar.** | **Yes** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Two triangles have A = D, AB = 3, AC = 5, DE = 6. What is DF if the triangles are similar? Student's answer: 6

**Use the scale factor from the matching sides.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The scale factor is 2. DF = 2 × 5 = 10.** | **10** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

a right triangle, the opposite side to angle A is 3 and the hypotenuse is 5. What is sin A? Student's answer: 4

**Use sin = opposite ÷ hypotenuse.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **4/5 is cos of the other angle. sin A = 3/5.** | **3/5** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

right triangle, the adjacent side to angle B is 8 and the hypotenuse is 17. What is cos B? Student's answer: 1

**Use cos = adjacent ÷ hypotenuse.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **15/17 uses the opposite side. cos B = 8/17.** | **8/17** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is tan 45°? Student's answer: $\sqrt{2}$

**Think about a 45-45-90 triangle.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **In a 45-45-90 triangle, the opposite and adjacent legs are equal, so tan 45° = 1.** | **1** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is sin 30°? Student's answer: $\sqrt{3}/2$

**Think about a 30-60-90 triangle.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **$\sqrt{3}/2$ is sin 60°. sin 30° = 1/2.** | **1/2** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A right triangle has opposite side 5 and adjacent side 12 for angle A. What is tan A? Student's answer: 12/5

**Use tan = opposite ÷ adjacent.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The student flipped the ratio. tan A = 5/12.** | **5/12** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is cos 60º? Student's answer: √3/2

**Think about a 30-60-90 triangle.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **√3/2 is cos 30°. cos 60° = 1/2.** | **1/2** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A right triangle has legs 3 and 4. What is the hypotenuse? Student's answer: 7

**Use the Pythagorean theorem.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The student added the legs. $3^2 + 4^2 = 25$, so the hypotenuse is 5.** | **5** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A right triangle has opposite side 5 and hypotenuse 10 for angle A. What is angle A? Student's answer: 60°

**Use sin A = opposite ÷ hypotenuse and an inverse sine.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **sin A = 5/10 = 1/2, so A = 30°.** | **30°** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A 10-foot ladder makes a 60° angle with the ground. How high does it reach? Student's answer: 5 feet

**Use the correct trig ratio.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The student used cosine. Height = 10 sin 60° = 5√3, about 8.66 feet.** | **5√3, about 8.66 feet** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A right triangle has angle 35° and hypotenuse 20. What is the adjacent side? Student's answer: 11.47

**Use cos = adjacent ÷ hypotenuse.**

What did the student do wrong? Why do you think the student made this error?

**The student used sine. The adjacent side is 20 cos 35°, about 16.38.**

Solve the problem correctly. Show all of your work.

**About 16.38**

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A right triangle has angle 50° and adjacent side 8. What is the opposite side? Student's answer: 6.72

**Use tan = opposite ÷ adjacent.**

What did the student do wrong? Why do you think the student made this error?

**The student divided by tan instead of multiplying. Opposite = 8 tan 50°, about 9.53.**

Solve the problem correctly. Show all of your work.

**About 9.53**

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

a building and looks up at the top, which is 20 feet high. The eye level is ignored. What is the angle of eleva

**Use tan = opposite ÷ adjacent.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The student used 15/20, which flips the ratio. tan  = 20/15, so  is about 53.13°.** | **About 53.13°** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

central angle measures 80°. What is the inscribed angle that intercepts the same arc? Student's answer: 8

**Recall that the inscribed angle is half the central angle.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **An inscribed angle is half the central angle that intercepts the same arc. 80 ÷ 2 = 40°.** | **40°** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

An inscribed angle measures 35°. What is the intercepted arc? Student's answer: 35°

**Recall that the arc is twice the inscribed angle.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The intercepted arc is twice the inscribed angle, so 2 × 35 = 70°.** | **70°** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

An inscribed angle is drawn with its sides on a diameter. What is its measure? Student's answer: 180°

**Think about the angle in a semicircle.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The intercepted arc is a semicircle, which is 180°. The inscribed angle is half of that, so it is 90°.** | **90°** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

angent line meets a radius at the point of tangency. What is the angle between them? Student's answer: 18

**Think about the tangent theorem.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **A tangent is perpendicular to the radius at the point of tangency, so the angle is 90°.** | **90°** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

rds intersect inside a circle. The intercepted arcs are 40° and 60°. What is the angle formed? Student's ans

**Use the intersecting chords theorem.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The angle equals half the sum of the arcs. (40 + 60) ÷ 2 = 50°.** | **50°** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A cyclic quadrilateral has an angle of 100°. What is the opposite angle? Student's answer: 100°

**Opposite angles in a cyclic quadrilateral are supplementary.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The opposite angles add to 180°, so the opposite angle is 80°.** | **80°** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the arc length of a 90° arc in a circle with radius 6. Student's answer: 6π

**Use arc length = (angle ÷ 360) × 2πr.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The student found the length of a full half circle. (90/360) × 12π = 3π, about 9.42.** | **3π, about 9.42** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the area of a 60° sector in a circle with radius 3. Student's answer: 9π

**Use sector area = (angle ÷ 360) × πr².**

What did the student do wrong? Why do you think the student made this error?

**The student found the area of the full circle. (60/360) × 9π = 1.5π, about 4.71.**

Solve the problem correctly. Show all of your work.

**1.5π, about 4.71**

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the arc length of a 120° arc in a circle with radius 9. Student's answer: 12π

**Use arc length = (angle ÷ 360) × 2πr.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The student used 2π × 9 × 2/3 incorrectly. (120/360) × 18π = 6π.** | **6π** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the area of a 180° sector in a circle with radius 4. Student's answer: 16π

**Use sector area = (angle ÷ 360) × πr².**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **16π is the area of the full circle. A half circle has area 8π.** | **8π** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the arc length of a 36° arc in a circle with radius 10. Student's answer: 3.6π

**Use arc length = (angle ÷ 360) × 2πr.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The student did not use the full circumference correctly. (36/360) × 20π = 2π, about 6.28.** | **2π, about 6.28** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A pizza of radius 7 is cut into 8 equal slices. What is the area of one slice? Student's answer: 49π/4

**Find the area of the full pizza, then divide by 8.**

What did the student do wrong? Why do you think the student made this error?

**49π/4 is the area of two slices. One slice is 49π/8, about 19.24.**

Solve the problem correctly. Show all of your work.

**49π/8, about 19.24**

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ctangle has vertices A(0, 0), B(4, 0), C(4, 3), and D(0, 3). What is the length of diagonal AC? Student's answ

**Use the distance formula.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The student added the side lengths. AC = $\sqrt{(4^2 + 3^2)} = 5$.** | **5** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is the slope of the line through A(1, 2) and B(5, 10)? Student's answer: 1/2

**Use (y - y) ÷ (x - x).**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The student flipped the ratio. The slope is 8 ÷ 4 = 2.** | **2** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Are the lines y = 2x + 1 and y = 2x - 3 parallel? Student says: No

**Compare slopes.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **Both lines have slope 2 and different intercepts, so they are parallel.** | **Yes** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Are lines with slopes 3 and -1/3 perpendicular? Student says: No

**Multiply the slopes.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **3 × (-1/3) = -1, so the lines are perpendicular.** | **Yes** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is the midpoint of (2, 5) and (8, -1)? Student's answer: (6, 3)

**Average the coordinates.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The student added the coordinates and then divided by 2 incorrectly. The midpoint is ((2 + 8)/2, (5 - 1)/2) = (5, 2).** | **(5, 2)** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is the triangle with vertices (0, 0), (4, 0), and (2, 3) isosceles? Student says: No

**Compare the side lengths.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The sides from (4, 0) to (2, 3) and from (2, 3) to (0, 0) are both $\sqrt{13}$ long. The triangle is isosceles.** | **Yes** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the midpoint of segment AB with A(0, 0) and B(10, 0). Student's answer: (4, 0)

**Use the midpoint formula.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The student made a subtraction error. The midpoint is (5, 0).** | **(5, 0)** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the point 1/3 of the way from A(0, 0) to B(12, 0). Student's answer: (3, 0)

**Multiply the x-distance by the fraction.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The student divided 12 by 4. 1/3 of 12 is 4, so the point is (4, 0).** | **(4, 0)** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the point that divides A(2, 0) to B(8, 0) in a 2:1 ratio, measured from A. Student's answer: (5, 0)

**Find 2/3 of the distance and add it to A.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The student used 1/2 of the distance. 2/3 of 6 is 4, so the point is (6, 0).** | **(6, 0)** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the midpoint of A(0, 0) and B(6, 9). Student's answer: (4.5, 3)

**Average the x-values and the y-values.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The student swapped the coordinates. The midpoint is (3, 4.5).** | **(3, 4.5)** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the point that is 3/4 of the way from (0, 0) to (8, 0). Student's answer: (2, 0)

**Multiply the distance by 3/4.**

What did the student do wrong? Why do you think the student made this error?

**The student used 1/4. 3/4 of 8 is 6, so the point is (6, 0).**

Solve the problem correctly. Show all of your work.

**(6, 0)**

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the midpoint of (1, 1) and (7, 7). Student's answer: (3, 4)

**Average each pair of coordinates.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The student made an error with the y-values. The midpoint is (4, 4).** | **(4, 4)** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A rectangle has vertices (0, 0), (5, 0), (5, 2), and (0, 2). What is its perimeter? Student's answer: 10

**Add all four sides.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The student counted each side once. The perimeter is 2(5 + 2) = 14.** | **14** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A right triangle has vertices (0, 0), (6, 0), and (0, 8). What is its area? Student's answer: 14

**Use base and height.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The student added the legs. Area = 1/2 × 6 × 8 = 24.** | **24** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A square has vertices (1, 1), (4, 1), (4, 4), and (1, 4). What is its area? Student's answer: 12

**Find the side length, then square it.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **12 is the perimeter. The side is 3, so the area is 9.** | **9** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A triangle has vertices (0, 0), (3, 4), and (6, 0). What is its area? Student's answer: 24

**Use base and height.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The student forgot the 1/2. Area = 1/2 × 6 × 4 = 12.** | **12** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A triangle has vertices (0, 0), (3, 4), and (6, 0). What is its perimeter? Student's answer: 14

**Find the length of each side with the distance formula.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The sides are 5, 5, and 6, so the perimeter is 16.** | **16** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A parallelogram has vertices (0, 0), (4, 0), (6, 3), and (2, 3). What is its area? Student's answer: 14

**Use base times height.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The student used the slanted side length. The height is 3, so the area is 4 × 3 = 12.** | **12** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the circumference of a circle with radius 4. Student's answer: 4π

**Use C = 2πr.**

What did the student do wrong? Why do you think the student made this error?

**The student forgot to multiply by 2. C = 2π(4) = 8π, about 25.13.**

Solve the problem correctly. Show all of your work.

**8π, about 25.13**

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the volume of a cylinder with radius 3 and height 5. Student's answer: 15π

**Use $V = \pi r^2 h$.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The student forgot to square the radius. $V = \pi(9)(5) = 45\pi$, about 141.37.** | **$45\pi$, about 141.37** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the volume of a cone with radius 3 and height 6. Student's answer: 54π

**Use $V = (1/3)\pi r^2 h$.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The student forgot the 1/3. $V = (1/3)\pi(9)(6) = 18\pi$, about 56.55.** | **$18\pi$, about 56.55** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the surface area of a sphere with radius 2. Student's answer: 8π

**Use $SA = 4\pi r^2$.**

What did the student do wrong? Why do you think the student made this error?

**The student forgot the 4 and used r instead of $r^2$. $SA = 4\pi(4) = 16\pi$, about 50.27.**

Solve the problem correctly. Show all of your work.

**$16\pi$, about 50.27**

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the surface area of a cylinder with radius 1 and height 2. Student's answer: $4\pi$

**Add the two circular bases and the side.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The student left out the bases. $SA = 2\pi(1)^2 + 2\pi(1)(2) = 6\pi$, about 18.85.** | **$6\pi$, about 18.85** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the area of a circle with radius 6. Student's answer: 12π

**Use $A = \pi r^2$.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The student doubled the radius instead of squaring it. $A = \pi(36) = 36\pi$, about 113.1.** | **$36\pi$, about 113.1** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A box is 4 by 5 by 6 units. What is its volume? Student's answer: 15

**Multiply all three dimensions.**

What did the student do wrong? Why do you think the student made this error?

**The student multiplied only two dimensions. V = 4 × 5 × 6 = 120.**

Solve the problem correctly. Show all of your work.

**120**

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A cylinder has volume 200π and radius 5. What is its height? Student's answer: 40

**Solve $V = \pi r^2 h$ for $h$.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The student divided by 5 instead of 25. $200\pi = \pi(25)h$, so $h = 8$.** | **8** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A sphere has volume 36π. What is its radius? Student's answer: 9

**Solve $V = (4/3)\pi r^3$ for r.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The student divided the volume by π and stopped. $36 = (4/3)r^3$ gives $r^3 = 27$, so $r = 3$.** | **3** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A cylindrical tank has radius 2 and height 10. What is its volume? Student's answer: 20π

**Use $V = \pi r^2 h$.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The student used r instead of $r^2$. $V = \pi(4)(10) = 40\pi$, about 125.66.** | **$40\pi$, about 125.66** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A cone has volume 16π and height 12. What is its radius? Student's answer: 4

**Solve $V = (1/3)\pi r^2 h$ for r.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **The student found $r^2 = 4$ and stopped. $r = \sqrt{4} = 2$.** | **2** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A prism has a base area of 15 square units and height 4 units. What is its volume? Student's answer: 19

**Use V = Bh.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The student added 15 and 4. V = 15 × 4 = 60.** | **60** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Which solid best models a soup can? Student's answer: cone

**Think about the shape of the can.**

What did the student do wrong? Why do you think the student made this error?

**A soup can has two circular bases and straight sides, so a cylinder is the best model.**

Solve the problem correctly. Show all of your work.

**Cylinder**

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Which solid best models a basketball? Student's answer: cylinder

**Think about the shape of the ball.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **A basketball is round in every direction, so a sphere is the best model.** | **Sphere** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ır garden is 12 feet by 7 feet and needs fencing around its edge. How much fencing is needed? Student's ar

**Find the perimeter of the rectangle.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The student added only one length and one width. Perimeter = 2(12 + 7) = 38 feet.** | **38 feet** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Which solid best models a traffic cone? Student's answer: cylinder

**Think about how the cone narrows.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **A traffic cone narrows to a point at the top, so a cone is the best model.** | **Cone** |

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A room is 10 feet by 12 feet. Carpet costs $4 per square foot. What is the total cost? Student's answer: $44

**Find the area, then multiply by the price.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
| --- | --- |
| **The student found the perimeter, which is 44 feet, instead of the area. Area = 120 square feet, so the cost is $480.** | **$480** |

Original task. Format inspired by error analysis routines.

# Find the Error



Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

pizza of radius 8 inches is cut into a slice with a 45° angle. What is the area of the slice? Student's answer: 6

**Find the area of the sector.**

| What did the student do wrong? Why do you think the student made this error? | Solve the problem correctly. Show all of your work. |
|---|---|
| **64π is the area of the full pizza. The slice area is (45/360) × 64π = 8π, about 25.13.** | **8π, about 25.13** |